UNITED **INTERNATIONAL UNIVERSITY (UIU)**

Dept. of Computer Science & Engineering

Class Test 5 (Set C)

Course No: CSE 4325 Course Title: Microprocessors and Microcontrollers

Time: 10 minutes **Marks: 20**

Name		**Section**		**ID**	

1. Write the **assembly code** for this expression using the **RAM content table** given below:

$$-1 + 2^2 \div \frac{1}{2} = ?$$

Then determine **the Accumulator and B register** values after executing each instruction. (10 + 10)

Initially, the value of **the Accumulator Register is 1, the B Register is 2.**

Address	Content		Address	Content
0H			8H	
1H			9H	
2H			AH	1000 0010
3H			BH	0000 0010
4H			CH	0100 0011
5H			DH	0010 0101
6H			EH	0001 0011
7H			FH	1100 0101

Solution:

Instruction	**AX**	**BX**
Initially	1	2
SUB BH	-1	2
ADD BH	1	2
ADD BH	3	2
ADD BH	5	2
ADD BH	7	2
OUT	7	2
HLT	0	0

UNITED **INTERNATIONAL UNIVERSITY (UIU)**

Dept. of Computer Science & Engineering

Class Test 5 (Set D)

Course No: CSE 4325 Course Title: Microprocessors and Microcontrollers

Time: 10 minutes **Marks: 20**

Name		**Section**		**ID**	

1. Write the **assembly code** for this expression using the **RAM content table** given below:

$$-1 \div \frac{1}{2} + 2^2 = ?$$

Then determine **the Accumulator and B register** values after executing each instruction. (10 + 10)

Initially, the value of **the Accumulator Register is 1, the B Register is 2.**

Address	Content		Address	Content
0H			8H	
1H			9H	
2H			AH	1100 0101
3H			BH	0000 0010
4H			CH	0100 0011
5H			DH	0010 0001
6H			EH	0001 0011
7H			FH	0000 0001

Solution:

Instruction	**AX**	**BX**
Initially	1	2
SUB BH	-1	2
SUB FH	-2	1
ADD BH	0	2
ADD BH	2	2
OUT	2	2
HLT	0	0